Raul Garrido Garcia

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Machine Learning / Applied Scientist | Computer Vision · Forecasting · Bayesian ML

4+ years building production-oriented deep-learning and ML systems for high-stakes decision support at scale, with model deployment on cloud infrastructure (AWS) and scalable, efficient ML pipelines. Ph.D. candidate in Physics (Northeastern, May 2027) and first-author in top-tier venues (Nature Communications; ARVO 2026). Strengths in computer vision, forecasting, Bayesian inference, and cross-functional product development. Open to Applied Scientist, Machine Learning Scientist, Data Scientist, Research Scientist, Computer Vision, and Quantitative Researcher roles across industries.

EDUCATION

Ph.D. in Physics, Northeastern University, MA, USA (expected May 2027) **09/2022 - Present**

MSc. in Physics, Northeastern University, MA, USA **09/2022 - 05/2024**

Bachelor of Science in Physics, Purdue University Northwest, IN, USA **08/2018 - 05/2022**

EXPERIENCE

Joslin Diabetes Center, Machine Learning / Applied Scientist Intern, Boston, MA **06/2026 - Present**

- Developed a computer-vision product for automated diabetic-retinopathy screening, aimed at commercialization with Optos (maker of the ultrawide-field retinal cameras used for screening), in cross-functional collaboration with clinicians.
- Built a production ML pipeline that combines multiple deep-learning models (PyTorch) for high precision at low inference latency, and re-engineered it to reduce memory footprint and runtime.
- Improved model performance through architecture optimization and hyperparameter tuning, and trained pigmentation-aware models specialized to distinct patient distributions to raise diagnostic accuracy across diverse populations and reduce reliance on manual clinician annotations.
- Supported model deployment to cloud infrastructure (AWS) for production use, in cross-functional collaboration with the engineering team.

Northeastern University, Research Assistant, Boston, MA **05/2023 - Present**

- Built and deployed a real-time Early Warning System for respiratory disease outbreak onsets and peaks across all 50 U.S. states (transfer learning) to help CDC and public-health officials assess outbreak risk earlier; detected 98% of onsets and 97% of peaks (5 and 2 weeks ahead on average) at 83% precision, narrowing detection windows from 10-20 weeks to 2-6 weeks.
- Developed an ensemble deep-learning computer-vision pipeline (PyTorch) to automate image-quality control in diabetic-retinopathy screening; evaluated on 27,000+ clinical ultrawide-field retinal images; presented at ARVO 2026.
- Built Bayesian / MCMC inference pipelines (NumPyro) to assess whether an epidemiological time series can be reliably fit by compartmental models (SIR/SEIR), and to quantify parameter uncertainty from population-level serological surveillance.
- Delivered reproducible analyses, visualizations, and ML decision-support reports for cross-functional engineering and clinical stakeholders (Python, SQL, Git).

PUBLICATIONS & PRESENTATIONS

- Real-time early-warning system for respiratory-disease outbreaks via transfer learning — Nature Communications (2026). doi.org/10.1038/s41467-026-72655-7
- Ensemble deep learning for retinal image-quality assessment in diabetic-retinopathy screening — ARVO 2026.
- Ethnicity-diverse training for equitable transformer-based diabetic-retinopathy detection — PLOS Digital Health (under review).
- Post-pandemic timing of influenza, RSV, and COVID-19 driving U.S. ILI — Lancet Regional Health – Americas (2024).

TECHNICAL SKILLS

Machine Learning & AI: Deep Learning | Computer Vision | Forecasting & Time-Series | Transfer Learning | Ensemble Methods | Bayesian Inference | Uncertainty Quantification | Model Evaluation

Programming Languages: Python | SQL | R | C++

ML Frameworks & Libraries: PyTorch | NumPyro | scikit-learn | Keras | TensorFlow | NumPy | SciPy | Pandas | matplotlib

Cloud & Deployment: AWS | Model Deployment | Production ML Pipelines | High-Performance Computing | Git

Collaboration & Communication: Cross-functional collaboration | Stakeholder communication | Technical writing | English (fluent) | Spanish (native)

LEADERSHIP, COMMUNICATION, AND MENTORSHIP

Instructor, Summer Institute in Statistics and Modeling in Infectious Diseases (SISMID) 07/2025

Halloran Scholarship to attend SISMID 07/2024

GLIAC Post Graduate Scholarship (sole conference-wide recipient) 05/2022

Co-president SPS and President Physics Club, Purdue University Northwest 08/2021 - 05/2022